1955 The U.S.S.R. explodes its first real superbomb, based on the Teller–Ulam/Sakharov–Zel'dovich idea. [Ch. 6]

Wheeler formulates the concept of gravitational vacuum fluctuations, identifies the Planck–Wheeler length as the scale on which they become huge, and suggests that on this scale the concept of spacetime gets replaced by quantum foam. [Ch. 12, 13, 14]

1957 Wheeler, Harrison, and Wakano formulate the concept of cold, dead matter and make a catalog of all possible cold, dead stars. Their catalog firms up the conclusion that massive stars must implode when they die. [Ch. 5]

Wheeler's group studies wormholes; Regge and Wheeler invent perturbation methods for analyzing small perturbations of wormholes; their formalism later will be used to study perturbations of black holes. [Ch. 7, 14]

Wheeler poses the issue of the final state of stellar implosion as a holy grail for research and, in a confrontation with Oppenheimer, opposes the idea that the final state will be hidden inside a black hole. [Ch. 6, 13J

1958 Finkelstein discovers a new reference frame for the Schwarzschild geometry, and it resolves the 1939 Oppenheimer–Snyder paradox of why an imploding star freezes at the critical circumference as seen from outside but implodes through the critical circumference as seen from inside. [Ch. 6]

1958–1960 Wheeler gradually embraces the concept of a black hole and becomes its leading proponent. [Ch. 6]

1959 Wheeler argues that spacetime singularities formed in the big crunch or inside a black hole are governed by the laws of quantum gravity, and may consist of quantum foam. [Ch. 13]

Burbidge shows that the giant lobes of radio galaxies contain magnetic and kinetic energy equivalent to that obtained by a perfect conversion of 10 million Suns into pure energy. [Ch. 9]